

PROGRAM 2 : STRING MANIPULATION

```
#include <iostream>
using namespace std;
void revStr(char* str) {
    int length = 0;
    while (str[length] != '\0') {
        length++;
    }
    for (int i = 0; i < length / 2; i++) {
        char temp = str[i];
        str[i] = str[length - i - 1];
        str[length - i - 1] = temp;
    }
}

void concatStr(char* s1, const char* s2) {
    int s1Length = 0;
    int s2Length = 0;
    while (s1[s1Length] != '\0') {
        s1Length++;
    }
    while (s2[s2Length] != '\0') {
        s1[s1Length + s2Length] = s2[s2Length];
        s2Length++;
    }
}
```

```
    s1[s1Length + s2Length] = '\0';  
}
```

```
void copyStr(char* s1, const char* s2) {  
    int i = 0;  
    while (s2[i] != '\0') {  
        s1[i] = s2[i];  
        i++;  
    }  
    s1[i] = '\0';  
}
```

```
int main() {  
    char str1[100], str2[100], str3[100];  
    cout << "Enter a string: ";  
    cin >> str1;  
    revStr(str1);  
    cout << "Reversed string: " << str1 << endl;  
  
    cout << "\nEnter two strings to concatenate: " << endl;  
    cout << "Enter string 1: ";  
    cin >> str1;  
    cout << "Enter string 2: ";  
    cin >> str2;  
    concatStr(str1, str2);  
}
```

```

cout << "Concatenated string: " << str1 << endl;
cout << "\nEnter a string to copy: ";
cin >> str2;
copyStr(str3, str2);
cout << "Copied string: " << str3 << endl;
return 0;
}

```

PROGRAM ! : MATRIX OPERATIONS

```

#include <iostream>
using namespace std;
class matrixOps {
private:
    int a[10][10];
    int b[10][10];
    int c[10][10];
    int x, y;

public:
    void get(int matrix[10][10]) {
        cout << "\nEnter elements of the matrix:" << endl;
        for (int i = 0; i < x; i++) {
            for (int j = 0; j < y; j++) {
                cout << "Enter element " << i + 1 << j + 1 << ": ";
                cin >> matrix[i][j];
            }
        }
    }
}

```

```
    }  
  }  
}
```

```
void display(int matrix[10][10]) {  
    for (int i = 0; i < x; i++) {  
        for (int j = 0; j < y; j++) {  
            cout << matrix[i][j] << "t";  
        }  
        cout << endl;  
    }  
}
```

```
void add() {  
    for (int i = 0; i < x; i++) {  
        for (int j = 0; j < y; j++) {  
            c[i][j] = a[i][j] + b[i][j];  
        }  
    }  
}
```

```
void subtract() {  
    for (int i = 0; i < x; i++) {  
        for (int j = 0; j < y; j++) {  
            c[i][j] = a[i][j] - b[i][j];  
        }  
    }  
}
```

```
}  
}
```

```
void multiply() {  
    for (int i = 0; i < x; i++) {  
        for (int j = 0; j < y; j++) {  
            c[i][j] = 0;  
            for (int k = 0; k < y; k++) {  
                c[i][j] += a[i][k] * b[k][j];  
            }  
        }  
    }  
}
```

```
void result() {  
    cout << "Enter number of rows: ";  
    cin >> x;  
  
    cout << "Enter number of columns: ";  
    cin >> y;  
  
    get(a);  
    get(b);  
  
    add();  
    cout << "Sum of matrices:" << endl;
```

```
    cout << "-----" << endl;
    display(c);
    subtract();
    cout << "Subtraction of matrices:" << endl;
    cout << "-----" << endl;
    display(c);
    multiply();
    cout << "Multiplication of matrices:" << endl;
    cout << "-----" << endl;
    display(c);
}
};

int main() {
    matrixOps m;
    m.result();
    return 0;
}
```